



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 1 • STANDARD 6.NOS.3

Multiply Decimals

Lesson 1-6 • Teacher Copy

Name: _____

Date: _____

Class: _____



TODAY'S OBJECTIVES

Content Objective: I can multiply decimals and place the decimal point correctly in the product.

Language Objective: I can explain my work using the words product, decimal point, estimate, and decimal places.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Product

Decimal point

Estimate

Place value

Decimal places



Tap any word to see what it means and a picture.

1

The answer to a multiplication problem is the .

2

The dot that separates whole numbers from parts of a whole is the

.

3

Finding an answer that is close but not exact, to check if a product is reasonable, is to

.

4

The value a digit has based on its position is its .

5

To place the point in a product, I count the total number of

in both factors.

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

EXPLAIN

When you multiplied 4.5×12.60 , how did you decide where to put the decimal point in the product?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

☐ **Product** — The answer when you multiply.
Producto — La respuesta cuando multiplicas.

☐ **Decimal point** — The dot that splits the whole number from the part after it.
Punto decimal — El punto que separa el número entero de la parte que sigue.

☐ **Estimate** — A close answer you get by rounding first.
Estimar — Una respuesta cercana que obtienes al redondear primero.

☐ **Place value** — What a digit is worth based on where it sits in a number.
Valor posicional — Lo que vale un dígito según el lugar que ocupa en el número.

☐ **Decimal places** — How many digits come after the decimal point.
Cifras decimales — Cuántos dígitos hay después del punto decimal.

Say it first: Say your idea to a partner or quietly to yourself before you write.

Di tu idea a un compañero o en voz baja antes de escribir.

I counted ____ decimal places in the factors.

Conté ____ lugares decimales en los factores.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: Rounding the factors gives a reasonable estimate of the product.

Evidence: Students round to about $5 \times \$13 = \65 (or $4 \times \$13$) to predict the cost is in the \$50-\$65 range, naming 4.5 and 12.60 as the factors.

Reasoning: This evidence connects the definition of product to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **I counted ____ decimal places in the factors.**

Conté ____ lugares decimales en los factores.

- **So my product has ____ decimal places, giving \$ ____.**

Entonces mi producto tiene ____ lugares decimales, dando \$ ____.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**

Mi afirmación es ____.

- **I know because ____.**

Lo sé porque ____.

- **So ____.**

Entonces ____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
Mi afirmación es ____.
- **My math evidence is ____.**
Mi evidencia matemática es ____.
- **This proves ____ because ____.**
Esto demuestra ____ porque ____.

4. Check Your Explanation

- ☐ I answered the question.
Respondí la pregunta.

- ☐ I used math evidence: numbers, an equation, a model, or a comparison.
Usé evidencia matemática: números, una ecuación, un modelo o una comparación.

- ☐ I explained how the evidence proves my answer.
Explicué cómo la evidencia demuestra mi respuesta.

- ☐ I used at least two math words.
Usé por lo menos dos palabras matemáticas.

- ☐ I reread complete sentences and punctuation.
Volví a leer mis oraciones completas y la puntuación.

Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

Answer Key & Teacher Guide

1. **Notes 1:** Product
2. **Notes 2:** Decimal point
3. **Notes 3:** Estimate
4. **Notes 4:** Place value
5. **Notes 5:** Decimal places
6. **Watch for this mistake:** *A common mistake in Multiply Decimals is counting decimal places from only one factor instead of adding the decimal places from BOTH factors. For example, in 0.4×0.06 , a student multiplies $4 \times 6 = 24$, then places the decimal point using just one factor's place count (1 place), writing 0.24 instead of the correct answer: 0.4 has 1 decimal place and 0.06 has 2 decimal places, for a total of 3, so 24 becomes 0.024. Before you submit, count the decimal places in EACH factor separately and add them together — that total is how many decimal places belong in your final answer.*
7. **Try It 1:** A. \$0.24 — $6 \times 4 = 24$. There are 2 total decimal places ($1 + 1$), so $0.6 \times 0.4 = 0.24$, which is \$0.24.
8. **Try It 2:** A. \$3.00 — $25 \times 12 = 300$. There are 2 total decimal places ($1 + 1$), so $2.5 \times 1.2 = 3.00 = \3.00 .